

Lista ②

① $\hat{A}$ e $\hat{B}$ $\alpha = \{\hat{e}_1, \hat{e}_2, \hat{e}_3, \hat{e}_4\}$

$$[\hat{A}]_\alpha = \frac{1}{2} \begin{pmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} \quad \text{e} \quad [\hat{B}]_\alpha = \frac{1}{4} \begin{pmatrix} 7 & -\sqrt{2} & 1 \\ -\sqrt{2} & 6 & \sqrt{2} \\ 1 & \sqrt{2} & 7 \end{pmatrix}$$

a)

$$[\hat{A}]_\alpha, [\hat{B}]_\alpha = \hat{A}\hat{B} - \hat{B}\hat{A}$$

$$= \frac{1}{8} \begin{pmatrix} 21+1 & -3\sqrt{2}+\sqrt{2} & 3+7 \\ -\sqrt{2} & 12 & 2\sqrt{2} \\ 7+3 & -\sqrt{2}+3\sqrt{2} & 1+21 \end{pmatrix} - \begin{pmatrix} 21+1 & -\sqrt{2} & 7+3 \\ -3\sqrt{2}+\sqrt{2} & 12 & -\sqrt{2}+3\sqrt{2} \\ 3+7 & \sqrt{2} & 1+21 \end{pmatrix}$$

$$[\hat{A}, \hat{B}] = \frac{1}{8} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = [0]$$

$$\bar{A} = 2A \text{ logo } \checkmark$$

b) $\det(\overbrace{[\hat{A}]_\alpha}^M - \lambda \mathbb{I}) = 0 \rightarrow \det\left(\frac{1}{2}(\bar{A} - 2\lambda \mathbb{I})\right)$

$$\det \begin{pmatrix} 3-2\lambda & 0 & 1 \\ 0 & 2-2\lambda & 0 \\ 1 & 0 & 3-2\lambda \end{pmatrix} = 0$$

$$\det(M) = \left(\begin{array}{ccc|cc} 3-2\lambda & 0 & 1 & 3-2\lambda & 0 \\ 0 & 2-2\lambda & 0 & 0 & 2-2\lambda \\ 1 & 0 & 3-2\lambda & 1 & 0 \end{array} \right) = (3-2\lambda)^2(2-2\lambda) - (2-2\lambda) = (2-2\lambda)[(3-2\lambda)^2 - 1] = 0$$

$$2-2\lambda=0 \quad z=2\lambda$$

$$(2-2\lambda)[(3-2\lambda)^2-1]=0$$

$$\lambda_1 = 1$$

$$\text{e de } (3-2\lambda)^2 = 1$$

$$9 - 12\lambda + 4\lambda^2 = 1$$

$$4\lambda^2 - 12\lambda + 8 = 0$$

$$\Delta = b^2 - 4ac$$

$$\Delta = 144 - 4 \cdot 4 \cdot 8$$

$$\Delta = 144 - 128$$

$$\Delta = 16 \Rightarrow \sqrt{\Delta} = 4$$

$$\text{então } \lambda_1 = \frac{12+4}{8} = 2 \quad \text{e } \lambda_2 = \frac{12-4}{8} = 1$$

$$\lambda_2 = 2$$

$$\lambda_3 = 1$$

$$\text{e } \lambda_1 = 1$$

Autovalores.

* Autoestados:

$$p/\lambda_1 = 1$$

$$[A] = \frac{1}{2} \begin{pmatrix} 3-2\lambda & 0 & 1 \\ 0 & 2-2\lambda & 0 \\ 1 & 0 & 3-2\lambda \end{pmatrix}$$

$$\Rightarrow [A]_2 = \begin{pmatrix} \frac{3}{2}-1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2}-1 \end{pmatrix}$$

$$\text{De } (A - \lambda I)|v\rangle = \vec{0}$$

$$\begin{pmatrix} \frac{3}{2}-1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2}-1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightsquigarrow \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow$$

$$\frac{1}{2}x + \frac{1}{2}z = 0 \Rightarrow z = -x$$

$$\frac{1}{2}x + \frac{1}{2}z = 0 \Rightarrow x = -z$$

e y é livre

2 autoestados pois $\lambda_1 = 1$
 $\lambda_3 = 1$

Assim, um autoestado é: $|\beta_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ pois $\langle \beta_1 | \beta_1 \rangle = 1$

e outro é $|\beta_3\rangle = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \frac{1}{\sqrt{2}}$ e $\langle \beta_3 | \beta_3 \rangle = 1$

o/ $\lambda_2 = 2$

$$\frac{1}{2} \begin{pmatrix} 3-2\lambda & 0 & 1 \\ 0 & 2-2\lambda & 0 \\ 1 & 0 & 3-2\lambda \end{pmatrix} = \begin{pmatrix} \frac{3-4}{2} & 0 & \frac{1}{2} \\ 0 & \frac{2-4}{2} & 0 \\ \frac{1}{2} & 0 & \frac{3-4}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix}$$

De $A|\beta_2\rangle = 2|\beta_2\rangle$

$$\begin{pmatrix} -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{cases} -\frac{1}{2}x + \frac{1}{2}z = 0 & (I) \\ -y = 0 & (II) \\ \frac{1}{2}x - \frac{1}{2}z = 0 & (III) \end{cases}$$

(II) mostra que $y = 0$

de (I): $-x + z = 0 \Rightarrow z = x$

de (III): $x - z = 0 \Rightarrow x = z$

então outro $|\beta_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

Portanto:

$$\beta = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} ; \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} ; \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

c) Placher $[B]_{\beta} = M^T B M$

$$M = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix} \rightarrow M^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 1 \end{pmatrix}$$

Logo:

$$[B]_{\beta} = \frac{1}{8} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7 & -\sqrt{2} & 1 \\ -\sqrt{2} & 6 & \sqrt{2} \\ 2 & \sqrt{2} & 7 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$= \frac{1}{8} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 6 & 8 \\ 6\sqrt{2} & -2\sqrt{2} & 0 \\ 2 & -6 & 8 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 12 & -4 & 0 \\ -4 & 12 & 0 \\ 0 & 0 & 16 \end{pmatrix}$$

d) $\det(B - \lambda I) = 0$

$$\begin{pmatrix} \frac{3}{2} - \lambda & -\frac{1}{2} & 0 & \frac{3}{2} - \lambda & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} - \lambda & 0 & -\frac{1}{2} & \frac{3}{2} - \lambda \\ 0 & 0 & 2 - \lambda & 0 & 0 \end{pmatrix} = \left(\frac{3}{2} - \lambda \right)^2 (2 - \lambda) - \frac{1}{4} (2 - \lambda) = 0$$

$$\left[\left(\frac{3}{2} - \lambda \right)^2 - \frac{1}{4} \right] (2 - \lambda) = 0$$

auto values

$$\begin{cases} \frac{9}{4} - 3\lambda + \lambda^2 - \frac{1}{4} = 0 \\ \lambda^2 - 3\lambda + 2 = 0 \end{cases} \quad \left\{ \begin{array}{l} D = 9 - 8 \\ \lambda_1 = \frac{3+1}{2} = 2 ; \lambda_2 = 1 \end{array} \right.$$

$\lambda_3 = 2 ; \lambda_1 = 2 ; \lambda_2 = 1$

Autoestados:

$$\begin{pmatrix} \frac{3}{2} - \lambda & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{3}{2} - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\lambda_1 = 1$$

$$\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{array}{l} \rightarrow \frac{1}{2}x - \frac{1}{2}y = 0 \quad (I) \\ -\frac{1}{2}x + \frac{1}{2}y = 0 \quad (II) \\ z = 0 \end{array}$$

De II $-x + y = 0 \Rightarrow \boxed{y = x}$ logo $|\lambda_1\rangle = \frac{1}{\sqrt{2}} (1, 1, 0)$

$$\lambda_2 = 2$$

$$\begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{array}{l} \rightarrow -\frac{1}{2}x - \frac{1}{2}y = 0 \quad (I) \\ -\frac{1}{2}x - \frac{1}{2}y = 0 \quad (II) \\ z = \text{livre} \end{array}$$

logo, De (II): $x = -y$ e z livre:

$$\mathcal{B} : \left\{ |\lambda_1\rangle = \frac{1}{\sqrt{2}} (1, -1, 0) \text{ e } |\lambda_2\rangle = (0, 0, 1) \quad |\lambda_3\rangle = \frac{1}{\sqrt{2}} (1, 1, 0) \right\}$$

Seja $|\lambda_a, \lambda_b\rangle \Rightarrow |1, 2\rangle ; |1, 1\rangle ; |2, 2\rangle$

$\hat{A}, \hat{B}$ formam um CCO.

$$e) M_{\mathcal{AB}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix} \quad \bigg| \quad M_{\text{out}} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$M_{\text{can}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \\ 1 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{pmatrix}$$

f) C, D atuam em V e $[C, D] = 0$

Seja M a matriz mudança de base. mostre que $[\bar{C}, \bar{D}] = 0$

$$\bar{C} = M^{-1} C M \text{ e } \bar{D} = M^{-1} D M$$

$$[M^{-1} C M, M^{-1} D M] = (M^{-1} C M)(M^{-1} D M) - (M^{-1} D M)(M^{-1} C M)$$

$$[\bar{C}, \bar{D}] = (M^{-1} C D M) - (M^{-1} D C M)$$

$$[\bar{C}, \bar{D}] = M^{-1} (C D - D C) M$$

$$[\bar{C}, \bar{D}] = M^{-1} [C, D] M = 0$$

g) já fiz na f)?

(2) a) $\text{Tr}(\hat{X}\hat{Y}) = \text{tr}(\hat{Y}\hat{X})$

$$\delta_{ij} \langle e_i | x y | e_j \rangle = \delta_{ij} \langle e_i | x | e_k \rangle \langle e_k | y | e_j \rangle$$

$$= \langle e_i | x | e_k \rangle \langle e_k | y | e_i \rangle$$

$$= \langle e_k | y | e_i \rangle \underbrace{\langle e_i | x | e_k \rangle}_1$$

$$= \langle e_k | y x | e_k \rangle$$

logo

$$\text{Tr}(\hat{X}\hat{Y}) = \text{Tr}(\hat{Y}\hat{X}).$$

$$b) (\hat{x}\hat{y})^+ = \hat{y}^+ \hat{x}^+$$

$$(\langle e_i | x y | e_j \rangle)^+ = \langle e_j | (xy)^+ | e_i \rangle$$

$$(\langle e_i | x | e_a \rangle \langle e_a | y | e_j \rangle)^+ = \langle e_j | (xy)^+ | e_i \rangle$$

$$\langle e_i | x | e_a \rangle^+ \langle e_a | y | e_j \rangle^+ = \langle e_j | (xy)^+ | e_i \rangle$$

$$\langle e_a | x^+ | e_i \rangle \langle e_j | y^+ | e_a \rangle = \langle e_j | (xy)^+ | e_i \rangle$$

$$\langle e_j | y^+ | e_a \rangle \underbrace{\langle e_a | x^+ | e_i \rangle}_{=1} = \langle e_j | (xy)^+ | e_i \rangle$$

$$\langle e_j | y^+ x^+ | e_i \rangle = \langle e_j | (xy)^+ | e_i \rangle$$

logo $\boxed{y^+ x^+ = (xy)^+}$